Q1

Calculate the intrinsic carrier density n_i for silicon at $T = 50$ K and 350 K.
Ans. $9.6 \times 10^{-39}/\text{cm}^3$; $4.15 \times 10^{11}/\text{cm}^3$

$$n_i = B \cdot T^{3/2} e^{-E_g/2kT}$$

$$B = 7.3 \times 10^{15}\ \text{cm}^{-3}\ K^{-3/2}$$

$$k = 8.62 \times 10^{-5}\ \text{eV/K}$$

$E_g = 1.12$ eV for Si

@ 50K → $n_i = (7.3 \times 10^{15})\, 50^{3/2} e^{-1.12/2k50}$

$= 9.63 \times 10^{-39}$ carriers/cm^3

@ 350K → $n_i = (7.3 \times 10^{15})\, 350^{3/2} e^{-1.12/2k350}$

$= 4.15 \times 10^{11}$ carrier/cm^3

Q2

For a silicon crystal doped with boron, what must N_A be if at $T = 300$ K the electron concentration drops below the intrinsic level by a factor of 10^6?

Ans. $N_A = 1.5 \times 10^{16}/\text{cm}^3$

$$n_e = n_i / 10^6$$

$$@\ T = 300K \Rightarrow n_i = 1.5 \times 10^{10}\ \text{carriers/cm}^3$$

$$n_i^2 = n_e \cdot n_p = n_e \cdot N_A$$

$$n_i^2 = \frac{n_i}{10^6} \cdot n_A \Rightarrow N_A = n_i \cdot 10^6 = 1.5 \times 10^{16}\ \text{cm}^{-3}$$